

RAJ PATEL

Vadodara, Gujarat

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Education

Parul Institute of Engineering and Technology

2022 – 2026

B.Tech – Computer Science and Engineering | CGPA: 7.42

Vadodara, India

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Machine Learning, Computer Networks, System Design

Technical Skills

- **Languages:** Python, C++, C, JavaScript, TypeScript, SQL
- **AI / ML:** LangGraph, LangChain, LLM Integration (Groq, Gemini, Llama), pgvector, HuggingFace Embeddings, NLP, Pandas, NumPy, Matplotlib
- **Backend / APIs:** FastAPI, Node.js, Express.js, REST API, WebSocket, JWT Authentication, asyncpg
- **Frontend:** React.js, Next.js, Three.js, Tailwind CSS, Redux / Redux Toolkit, Zustand, Framer Motion
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis, SQLAlchemy ORM, pgvector
- **DevOps / Tools:** Docker, Git, GitHub, VS Code, PyCharm, Postman, Vercel, Render
- **External APIs:** Tavily Search, Google Serper, Groq, Google Gemini, TMDB

Projects

NexusAI — Production-Grade Multi-Agent AI Platform | [GitHub](#)

Apr 2025 – Jun 2025

Next.js 16, React 19, TypeScript, Python, FastAPI, LangGraph, LangChain, Groq (Llama 3.3-70B), PostgreSQL, pgvector, Redis, Docker, Zustand, WebSocket

Built a production-grade multi-agent AI platform to automate complex coding workflows — from research and code generation to debugging and review — with real-time execution and Human-in-the-Loop control.

- Architected a 6-agent LangGraph pipeline (Router → Search → Coder → Debugger → Critic → Writer) with conditional routing, real-time WebSocket token streaming, and a HITL approval gate using LangGraph interrupt_before.
- Implemented Docker sandboxed code execution (--cap-drop ALL, --network none, 10s timeout) supporting 20+ languages, backed by a two-tier semantic cache (Redis + pgvector HNSW) with Jaccard IoU anti-hallucination guard to eliminate redundant LLM API calls.

RAG Document Intelligence — Production-Grade RAG API Platform | [GitHub](#)

Feb 2025 – Mar 2025

Python, FastAPI, LangChain, PostgreSQL, pgvector, MongoDB Atlas, HuggingFace Embeddings, asyncpg, Docker, JWT

Developed a scalable document intelligence API to enable accurate semantic search over user-uploaded files, solving the problem of context bleed across documents in shared vector indexes.

- Engineered ID-based retrieval with HNSW indexing and dual vector store support (PostgreSQL pgvector + MongoDB Atlas) via a factory pattern, enabling per-file semantic search with multi-tenant schema isolation.
- Built an async producer-consumer batch embedding pipeline (asyncio.Queue, batches of 500) with a pluggable provider system supporting OpenAI, HuggingFace, Gemini, and AWS Bedrock — switchable via environment variables with zero code changes.

AI Resume & Job Matcher — Full-Stack AI Job Search Platform | [GitHub](#)

Jul 2025 – Aug 2025

Python, FastAPI, React.js, NLP, TF-IDF, JobSpy, Google Gemini API, PostgreSQL

Built a full-stack AI platform that automates job discovery by parsing resumes, extracting skills with NLP, and matching candidates to live job listings — eliminating manual keyword-based job searches.

- Designed an NLP pipeline (spaCy + TF-IDF) to extract skills from uploaded resumes and auto-generate optimized job board queries, then scraped live listings from LinkedIn, Indeed, and Glassdoor in real-time via JobSpy.
- Implemented cosine similarity scoring between resume embeddings and job descriptions to rank best-matched opportunities with match percentage, persisted via a FastAPI + PostgreSQL backend with a React.js frontend.

Certifications

- Certified in Computer Networks & Protocols — NPTEL Swayam Portal (Score: 78%)
- Certified in Introduction to Internet of Things — NPTEL Swayam Portal (Score: 70%)